

ClaimSwift - Official Inter-Service API Contract Document

1. Overview

This document defines the official internal API contracts between microservices for the ClaimSwift Insurance Claims Processing Platform.

All services must:

- Validate JWT tokens locally.
- Use ROLE_SERVICE for internal communication.
- Avoid direct database access to other services.
- Expose internal endpoints under /internal/**.

2. Claim Service Contract (Required by Reporting & Notification)

Endpoint: GET /internal/claims/summary

Authorization: Bearer <SERVICE_JWT>

Response:

```
{
  "total": Long,
  "approved": Long,
  "rejected": Long,
  "avgSettlementTime": Double
}
```

Optional Endpoint: GET /internal/claims/adjuster-performance

Response:

```
[
  {
    "adjusterId": Long,
    "claimsProcessed": Long,
    "avgProcessingTime": Double
  }
]
```

3. Payment Service Contract (Required by Reporting)

Endpoint: GET /internal/payments/summary

Authorization: Bearer <SERVICE_JWT>

Response:

```
{
  "totalPayments": Long,
  "totalAmountSettled": Double,
  "avgSettlementProcessingTime": Double
}
```

4. Notification Service Contract

Endpoint: POST /internal/notifications/event

Authorization: Bearer <SERVICE_JWT>

Request Body:

```
{
  "userId": Long,
  "claimId": Long,
  "title": String,
  "message": String
}
```

Response: 200 OK

5. Security Contract

JWT Requirements:

- Issued only by Auth Service
- Must contain 'role' claim
- Roles: POLICYHOLDER, ADJUSTER, MANAGER, SERVICE
- Expiry must be validated by each microservice

Internal Calls:

- Must include Authorization header
- Must use ROLE_SERVICE

6. Standard Error Response Format

```
{
  "status": 400,
  "message": "Error description",
  "timestamp": "2026-01-01T12:00:00"
}
```

7. Team Responsibilities for Integration

- Auth Team: Provide JWT with correct role claim.
- Claim Team: Implement required /internal endpoints.

- Payment Team: Provide payment aggregation endpoint.
- Notification Team: Accept internal event API.
- Reporting Team: Aggregate data without direct DB joins.
- Infrastructure Team: Maintain Eureka & API Gateway.